

# Python for Research

## An Initiation

The purpose of research is to try and explain something to the world based on certain observations. The challenge is to come up with an explanation before anyone else does. You, as a researcher, are confronted with a volume of data. You need to model it and demonstrate that your model is correct.

How likely are you to hit upon a valid model on your first try? Not very, unless the problem is trivial. So, the challenge is not just to prove that the correct model works. The even greater challenge is to find out as quickly as possible, whether your model is the correct solution, and if not, to eliminate it and move on. This is precisely where Python is superb.

Well, there are some issues.

Python works with dynamic types. A list can contain any mixture of data types. The research problem deals with data of one type. The mixture of data types, obviously, allows for the data to be of the same type as well. However, there is a price. The type of data has to be checked at run time. As a consequence, for massive computation, Python can be virtually unusable.

So, Jim Hugunin, a graduate student at MIT, created Numeric in 1995. He then went on to create Jython and IronPython. Numeric continued to grow in the scientific and engineering community. Many students and scientists built their modules around the Numeric module. The project's name on SourceForge.net was NumPy. This

Let's get started with NumPy, which is supposed to be "the fundamental package needed for scientific computing with Python."

